



Modelling Financial Markets

Seminar 20: Path Signatures and Realised Volatility — Hands-on



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ AUDUSD: Australian dollar / US dollar exchange rate (FX pair code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ CI: Confidence Interval
- ▣ CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CVX: Chevron Corporation (NYSE ticker)
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: E-mini S&P 500 futures (CME ticker)
- ▣ EURUSD: Euro / US dollar exchange rate (FX pair code)
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ GBPUSD: British pound / US dollar exchange rate (FX pair code)
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ OLS: Ordinary Least Squares
- ▣ OPEC: Organization of the Petroleum Exporting Countries
- ▣ PRDS: Positive Regression Dependence on a Subset (of p-values)
- ▣ QLIKE: Quasi-LIKElihood (loss function)

Acronyms used in this material (2/2)

- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SHAR: Semivariance HAR model
- ▣ SHW: Sherwin-Williams Company (NYSE ticker)
- ▣ US: United States
- ▣ USDCAD: US dollar / Canadian dollar exchange rate (FX pair code)
- ▣ USDJPY: US dollar / Japanese yen exchange rate (FX pair code)
- ▣ VOLARE: VOLatility Archive for Realized Estimates (database)

Today's Questions and Route

- ▣ Questions of the seminar
 - ▶ does the shape of the recent volatility path help to forecast realised variance?
 - ▶ how do we test that honestly?
- ▣ Route
 - ▶ First the primer “What You Need for Today” (the lecture comes after the seminar); then A1, A7, A9 on paper: a signature step by step, the QLIKE loss, kernel weights
 - ▶ B1, B3 on JPM: the Lévy area as a predictor and the HAR family, with HAC errors
 - ▶ B5: Sig-LK against log-HAR with DM, GW and a bootstrap interval; B7: Holm and BH over 50 assets
 - ▶ C2: find the errors in an AI answer
- ▣ What you produce: the A answers, one figure and one table for each of B1, B3, B5, B7, the corrected C2 code, one interpretation sentence per exercise
- ▣ Notebook for today: [open the seminar notebook in Colab](#); background: the primer below, [Chapter 9](#); [Chapter 20](#) follows after the seminar
- ▣ Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A1	how does a signature record the order of moves?	Solved	–
A7	which forecast minimises QLIKE, which errors cost more?	Solved	–
A9	how does the temperature trade similarity for sample size?	Solved	–
B1	does the Lévy area predict next month's RV of JPM?	Solved	–
B3	do the SHAR and HARQ effects hold for JPM?	Solved	–
B5	does Sig-LK beat log-HAR for JPM, one day ahead?	Solved	–
B7	how many wins survive Holm and BH over 50 assets?	Solved	–
C2	what is wrong in an AI answer?	Proposed	A1, A7, B5
Extension packs	A2–A6, A8, A10, B2, B4, B6, B8, C1 (optional, after the lecture)	Solved and Proposed (marked on each task)	named on each task

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: verification and inference on VOLARE data, each ending with an interpretation question; Part C: open questions and AI critique

Data Used

Data set	Assets	Sample	Used in
VOLARE realised measures, US stocks	JPM (all 40 stocks in C1)	2 Jan 2015 – 30 Jan 2026	A5, B1, B3, B5, C1
VOLARE realised measures, futures	ES (E-mini S&P 500), CL (crude oil)	28 Sep 2009 – 30 Jan 2026	B2, B4, B6
Lecture 20 results (provided with the notebook)	50 assets	one-sided DM p -values	B7, B8

- Variables of the VOLARE data [Cipollini et al. \(2026\)](#), one value per asset and trading day
 - ▶ Realised variance from five-minute returns, $RV_t = \sum_i r_{t,i}^2$, and its positive and negative parts (semivariances RV_t^+ , RV_t^-)
 - ▶ Realised quarticity (RQ), the scale of the measurement error of RV; the daily open and close prices, for the open-to-close log return r_t
- VOLARE is free for research at volare.unime.it: download the realised-variance data (stocks, futures, FX) and point the first notebook cell to them

Setup: Run and Verify

```
# notebook section "Signatures, models and tests"
# defines load_asset(kind, symbol): columns
# rv, rsp, rsn, rq and r = ln(close/open)
a = load_asset('stocks', 'JPM')
print(len(a), a.index[0].date(), a.index[-1].date())
print(pd.DataFrame({'RV_x_1e4': 1e4 * a.rv,
                    'r(%)': 100 * a.r}).head(3))
```

- Expected output: 2786 trading days, 2015-01-02 to 2026-01-30

date	RV $\times 10^4$	r (%)
2015-01-02	1.463	-0.22
2015-01-05	1.855	-2.56
2015-01-06	2.974	-2.77

- RV of 1.46×10^{-4} is a daily volatility of $\sqrt{1.46 \times 10^{-4}} = 1.2\%$
- If your count or first value differs, check the data set and the asset

A Window of Data Is a Path; Its Signature

- The seminar comes before the lecture: these frames give only what today's tasks need; theorems, the full design and the 50-asset results in [Chapter 20](#), after the seminar
- Take the last L days of a few series and join the points by straight lines: a path $X : [a, b] \rightarrow \mathbb{R}^d$
 - ▶ **Time augmentation:** channel 1 is time itself, e.g. $X = (t, \ln RV_t, \sum_{s \leq t} r_s)$; without it a fast move and a slow move look the same
- **Signature:** all iterated integrals $S^{i_1 \dots i_k} = \int_{a < u_1 < \dots < u_k < b} dX_{u_1}^{i_1} \dots dX_{u_k}^{i_k}$, collected by level k [Chevyrev & Kormilitzin \(2016\)](#)
 - ▶ Level 1: $S^i = X_b^i - X_a^i$, the total increments; level 2: $S^{ij} = \int_a^b (X_u^i - X_a^i) dX_u^j$, a $d \times d$ matrix
 - ▶ One straight segment with increment Δ : level 1 is Δ , level 2 is $\frac{1}{2} \Delta \Delta^\top$
- Why use it: with a time channel the signature determines the path [Hambly & Lyons \(2010\)](#), and linear functions of the truncated signature approximate continuous functions of the path, so it serves as a set of regression features

Chen's Identity, the Shuffle Identity and the Lévy Area

- **Chen's identity** [Chen \(1957\)](#): the signature of two joined paths is the tensor product of their signatures; at level 2
 - ▶ $S^{(2)}(X * Y) = S^{(2)}(X) + S^{(1)}(X) S^{(1)}(Y)^\top + S^{(2)}(Y)$
 - ▶ Applied segment by segment: $S^{(2)} = \sum_k \frac{1}{2} \Delta_k \Delta_k^\top + \sum_{j < k} \Delta_j \Delta_k^\top$
- **Shuffle identity**: $S^i S^j = S^{ij} + S^{ji}$; so the symmetric part of level 2 is $\frac{1}{2} \Delta X^i \Delta X^j$ and brings nothing new
- **Lévy area** $A^{ij} = \frac{1}{2} (S^{ij} - S^{ji})$: the signed area between the path and its chord; it records the **order** of the moves
- Small example, channels (time, y), $y = (0, 1, 1)$: $\Delta_1 = (1, 1)$, $\Delta_2 = (1, 0)$
 - ▶ $S^{(1)} = (2, 1)$; $S^{(2)} = \frac{1}{2} \Delta_1 \Delta_1^\top + \frac{1}{2} \Delta_2 \Delta_2^\top + \Delta_1 \Delta_2^\top = \begin{bmatrix} 2 & 0.5 \\ 1.5 & 0.5 \end{bmatrix}$; shuffle: $S^1 S^2 = 2 = 0.5 + 1.5$
 - ▶ $A^{12} = (0.5 - 1.5)/2 = -0.5$: y rose early; for $y = (0, 0, 1)$ (late rise) the same steps give $A^{12} = +0.5$

Sig-LK: the Forecasting Method Tested in Chapter 20

- Signature LASSO with kernel weights of [Gu et al. \(2024\)](#), applied to next-day (or next-week, next-month) log RV; benchmark: log-HAR ([Chapter 9](#))
- Features ξ_τ : the signature up to depth 3 of the 22-day path (time, ln RV, cumulative r) ending at day τ
- Kernel weights: $\delta_\tau =$ squared distance between the truncated signatures of a past window and of the latest window;
 $w_\tau = e^{-\gamma\delta_\tau} / \sum_s e^{-\gamma\delta_s}$
 - ▶ Temperature $\gamma = c/\text{median}(\delta)$: with $c = 1$ a window at the median distance gets weight $e^{-1} = 0.37$ relative to an identical one; $\gamma = 0$ gives equal weights
 - ▶ Cost: fewer effective observations, Kish $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$ (A9)
- Two steps: a weighted LASSO selects the features (λ chosen by BIC), then weighted OLS refits the selected ones, removing the LASSO shrinkage
- Rolling design: the last 1000 complete pairs (features, target), re-estimated every 5 days; each forecast uses only data up to its origin (no look-ahead)

Tools from Earlier Chapters, Task by Task

- B1, B3: HAR and log-HAR [Corsi \(2009\)](#), SHAR with semivariances [Patton & Sheppard \(2015\)](#), HARQ with realised quarticity [Bollerslev, Patton & Quaedvlieg \(2016\)](#), Newey–West (HAC) standard errors: [Chapter 9](#)
- A7, B5: QLIKE and why it ranks variance forecasts robustly to a noisy proxy [Patton \(2011\)](#): [Chapter 9](#)
- B5: Diebold–Mariano [Diebold & Mariano \(1995\)](#) and conditional Giacomini–White [Giacomini & White \(2006\)](#) tests: [Chapter 8](#); stationary bootstrap for an interval: [Chapter 5](#)
- B7: Holm (FWER) and Benjamini–Hochberg (FDR) corrections [Benjamini & Hochberg \(1995\)](#): [Chapter 3](#)
- New today are only the signature, the Lévy area and the kernel weights of the three previous slides

Notation Used in Every Exercise

- ▣ RV_t : five-minute realised variance of day t ; $r_t = \ln(\text{close}_t / \text{open}_t)$: open-to-close log return
- ▣ HAR averages: $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$, $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$; log-HAR regresses $\ln RV_{t+1:t+h}$ on $\ln RV_t$, $\ln RV_t^{(w)}$, $\ln RV_t^{(m)}$, with $RV_{t+1:t+h}$ the mean RV over the next h days
- ▣ Path X with increments Δ_k ; signature levels $S^{(1)} = \sum_k \Delta_k$ and $S^{(2)}$ (a $d \times d$ matrix); Lévy area $A^{12} = (S^{12} - S^{21})/2$
- ▣ Loss $\text{QLIKE}(RV, F) = RV/F - \ln(RV/F) - 1$ for a forecast F ; loss difference $d_t = \text{QLIKE}_t^{(1)} - \text{QLIKE}_t^{(2)}$; DM statistic $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}}/T}$ with a Newey–West (HAC) long-run variance
- ▣ Kernel weights $w_\tau = e^{-\gamma \delta_\tau} / \sum_s e^{-\gamma \delta_s}$ for signature distances δ_τ ; Kish effective sample size $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- ▣ Worked example: $RV = 1$, $F = 2$: $\text{QLIKE} = 0.5 - \ln 0.5 - 1 = 0.193$

A1 [Solved]: A Depth-2 Signature, Step by Step — Task

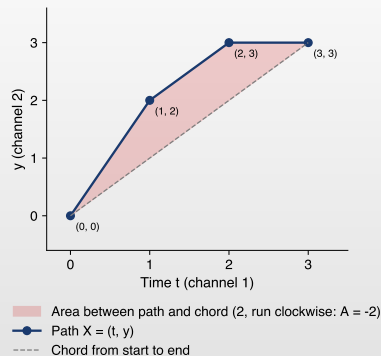
- **Question:** how does a depth-2 signature record *when* a series moved, not only how much?
- **Inputs**
 - ▶ A series takes the values $y = (0, 2, 3, 3)$ on days $t = 0, 1, 2, 3$; no data file
 - ▶ The time-augmented path $X_t = (t, y_t)$ is linear between consecutive days (channel 1 = time, channel 2 = y)
- **Tasks**
 - (a) Write the three increments $\Delta_1, \Delta_2, \Delta_3 \in \mathbb{R}^2$ and compute level 1, $S^{(1)} = \sum_k \Delta_k$.
 - (b) Use Chen's identity to show that $S^{(2)} = \sum_k \frac{1}{2} \Delta_k \Delta_k^\top + \sum_{j < k} \Delta_j \Delta_k^\top$, then compute this 2×2 matrix.
 - (c) Check the result with the shuffle identity $S^i S^j = S^{ij} + S^{ji}$ for $i = j = 1$ and for $i = 1, j = 2$.
 - (d) Compute the Lévy area $A^{12} = (S^{12} - S^{21})/2$ and draw the path with its chord.
- **Report:** $S^{(1)}$, $S^{(2)}$ and A^{12}
- **Interpretation:** what does the sign of A^{12} say about when the series rose?
- Notebook: section A1

A1 [Solved]: Solution (1/2) — Levels 1 and 2

- (a) Increments: $\Delta_1 = (1, 2)$, $\Delta_2 = (1, 1)$, $\Delta_3 = (1, 0)$; level 1: $S^{(1)} = (3, 3)$, the total change in time and in y
- (b) One segment has $\text{Sig} = (1, \Delta, \frac{1}{2}\Delta\Delta^\top, \dots)$; Chen multiplies the segments, and level 2 of the product collects every square term once and every ordered pair $j < k$ once
- Squares: $\frac{1}{2} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$
- Ordered pairs: $\Delta_1\Delta_2^\top + \Delta_1\Delta_3^\top + \Delta_2\Delta_3^\top = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$
- Sum: $S^{(2)} = \begin{bmatrix} 4.5 & 2.5 \\ 6.5 & 4.5 \end{bmatrix}$

A1 [Solved]: Solution (2/2) — Shuffle Check and Lévy Area

- (c) Shuffle, $i = j = 1$: $S^1 S^1 = 9 = 2S^{11} = 2 \cdot 4.5$; $i = 1, j = 2$: $S^1 S^2 = 9 = S^{12} + S^{21} = 2.5 + 6.5$: both hold
- (d) Lévy area: $A^{12} = (2.5 - 6.5)/2 = -2$
- Geometric check (chart): the path lies above its chord $y = t$; the shaded polygon has area 2 and is run clockwise, hence the minus sign
- Interpretation: with channels (time, y), a negative area means that y rose *early* in the window; the same net change spread evenly would give $A^{12} = 0$



A7 [Solved]: QLIKE: Minimiser and Asymmetry — Task

□ Questions

- ▶ which forecast minimises expected QLIKE?
- ▶ which errors does QLIKE punish more?

- **Inputs:** $\text{QLIKE}(RV, F) = RV/F - \ln(RV/F) - 1$ with realised proxy $RV > 0$ and forecast $F > 0$; $\mathbb{E}_t =$ expectation given the information at the forecast origin

□ Tasks

- Differentiate $\mathbb{E}_t[\text{QLIKE}(RV, F)]$ with respect to F and solve the first-order condition.
- Check the second-order condition.
- For $RV = 1$, compute QLIKE and the squared error for $F = 0.5$ and for $F = 1.5$, and plot both losses against F .

- **Report:** the minimiser, the four losses and the chart

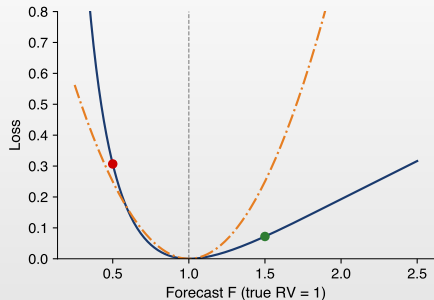
□ Interpretation

- ▶ which error does QLIKE punish more?
- ▶ why does this suit risk management?

- Notebook: section A7

A7 [Solved]: Solution

- (a) $\mathbb{E}_t[\text{QLIKE}] = \mathbb{E}_t[\text{RV}]/F + \ln F - \mathbb{E}_t[\ln \text{RV}] - 1$;
 $\partial/\partial F = -\mathbb{E}_t[\text{RV}]/F^2 + 1/F = 0 \Rightarrow F^* = \mathbb{E}_t[\text{RV}]$
- (b) $\partial^2/\partial F^2 = 2\mathbb{E}_t[\text{RV}]/F^3 - 1/F^2 = 1/F^{*2} > 0$ at F^* : a minimum; F^* depends only on $\mathbb{E}_t[\text{RV}]$, so with a conditionally unbiased proxy the *expected-loss* ranking is robust [Patton \(2011\)](#)
- (c) $\text{QLIKE}(1, 0.5) = 2 - \ln 2 - 1 = 0.307$;
 $\text{QLIKE}(1, 1.5) = 0.667 + \ln 1.5 - 1 = 0.072$; squared errors 0.25 and 0.25
- Interpretation: QLIKE punishes under-prediction of variance about four times more than the same over-prediction; under-predicted risk is the costly error (too little capital, breached VaR)



— QLIKE(RV = 1, F)

- - - Squared error $(1 - F)^2$

● F = 0.5: QLIKE 0.307

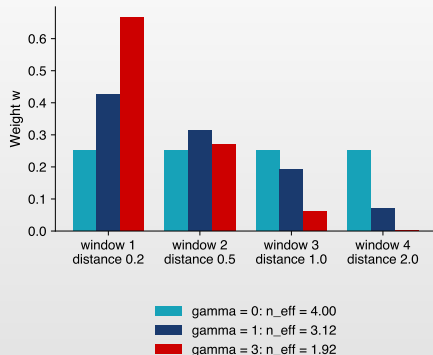
● F = 1.5: QLIKE 0.072

A9 [Solved]: Kernel Weights and Effective Sample Size — Task

- **Question:** how does the kernel temperature γ trade similarity against the number of observations?
- **Inputs**
 - ▶ Four past windows with signature distances $\delta = (0.2, 0.5, 1.0, 2.0)$ to the current window
 - ▶ Weights of Gu et al. (Eq. 12): $w_\tau = e^{-\gamma\delta_\tau} / \sum_s e^{-\gamma\delta_s}$; Kish effective sample size $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- **Tasks**
 - (a) Compute the four weights for $\gamma = 1$.
 - (b) Compute the four weights for $\gamma = 3$.
 - (c) Compute n_{eff} in both cases and compare with equal weights ($\gamma = 0$).
- **Report:** two weight vectors, two n_{eff} , a bar chart
- **Interpretation**
 - ▶ what is gained when γ increases?
 - ▶ what is lost?
- **Notebook:** section A9

A9 [Solved]: Solution

- (a) $e^{-\delta} = (0.819, 0.607, 0.368, 0.135)$, sum 1.929:
 $w = (0.425, 0.315, 0.191, 0.070)$
- (b) $e^{-3\delta} = (0.549, 0.223, 0.050, 0.002)$, sum 0.824:
 $w = (0.666, 0.271, 0.060, 0.003)$
- (c) $n_{\text{eff}} = 4$ for $\gamma = 0$, 3.12 for $\gamma = 1$ and 1.92 for $\gamma = 3$
- Interpretation: a higher temperature focuses on the most similar window (less bias from dissimilar periods) but halves the information (more variance); in the lecture $n_{\text{eff}} \approx 816$ of 1000: little is lost, and little changes



B1 [Solved]: Does the Lévy Area Predict Volatility? (JPM) — Task

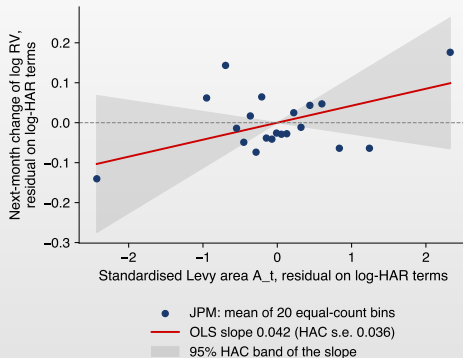
- **Question:** does the order of price and volatility moves over the last month predict JPM's volatility next month, beyond log-HAR?
- **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; five-minute RV, daily open and close prices
 - ▶ Regressor: $A_t = \text{Lévy area of the 22-day path (cumulative } r, \ln RV) \text{ ending at day } t, \text{ divided by its standard deviation}$
 - ▶ Target: $y_t = \ln RV_{t+1:t+22} - \ln RV_t^{(m)}$, next month's log RV minus the past month's (consecutive targets overlap in 21 of 22 days)
- **Tasks**
 - (a) For every day, build the path of the last 22 days, compute its depth-2 signature and $A_t = (S^{12} - S^{21})/2$, then standardise A_t .
 - (b) Regress y_t by OLS on a constant, the three log-HAR terms and A_t , with Newey–West standard errors and 22 lags.
 - (c) Test H_0 : coefficient of $A_t = 0$ at 5%, two-sided, and compare the R^2 with and without A_t .
 - (d) Draw a binned scatter of y_t against A_t , both residualised on the log-HAR terms.
- **Report:** coefficient, HAC s.e., t , decision, both R^2 , the chart
- **Interpretation**
 - ▶ why 22 lags?
 - ▶ is the gain economically relevant?
- **Notebook:** section B1

B1 [Solved]: Solution (1/2) — Steps

- (a) Signature of each 22-point window with channels (return, $\ln RV$); A_t is negative in 46% of the windows (compatible with, but not identifying, leverage-type sequences)
- (b) 2743 usable days; coefficient of $A_t = +0.042$, HAC s.e. 0.036
- (c) $t = +0.042/0.036 = +1.18$; $|t| < 1.96$: do not reject H_0 at 5%; R^2 : 28.0% without A_t , 28.6% with A_t
- Why 22 lags: consecutive targets share 21 of their 22 days, so the errors are serially dependent even under H_0 ; persistent RV can extend the dependence, so a check with 44 lags is advisable

B1 [Solved]: Solution (2/2) — Chart and Interpretation

- (d) Chart: bins scatter around zero; the HAC band of the slope contains the horizontal line
- Relevance: half a point of R^2 next to the 28% explained by the log-HAR terms is negligible
- Interpretation: for JPM the order of moves between price and volatility adds no reliable information about next month's volatility



B3 [Solved]: HAR, SHAR and HARQ for JPM — Task

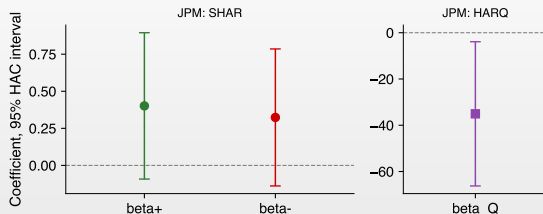
- **Question:** do bad volatility (SHAR) and the measurement error of RV (HARQ) change next-day forecasts for JPM?
- **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; five-minute RV, its semivariances RV_t^+ , RV_t^- and realised quarticity RQ; target: next-day RV in levels, full sample
 - ▶ SHAR [Patton & Sheppard \(2015\)](#): $RV_{t+1} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$
 - ▶ HARQ [Bollerslev, Patton & Quaedvlieg \(2016\)](#): HAR plus the term $\beta_Q RV_t (RQ_t^{1/2} - \overline{RQ^{1/2}})$; V = Newey–West (5 lags) covariance matrix of the estimates
- **Tasks**
 - (a) Estimate HAR [Corsi \(2009\)](#) by OLS and report the three slopes with HAC standard errors.
 - (b) Estimate SHAR and test $H_0 : \beta^- = \beta^+$ against $\beta^- > \beta^+$ with $t = (\hat{\beta}^- - \hat{\beta}^+) / \sqrt{V_{--} + V_{++} - 2V_{+-}}$.
 - (c) Estimate HARQ and test $H_0 : \beta_Q = 0$ against $\beta_Q < 0$; plot β^+ , β^- and β_Q with 95% HAC intervals.
- **Report:** estimates, t , p , decisions, the coefficient chart
- **Interpretation:** do the two effects of the literature hold for this stock?
- **Notebook:** section B3

B3 [Solved]: Solution (1/2) — Estimates and Tests

- (a) HAR (2 764 days): $\beta_d = 0.367$ (0.080), $\beta_w = 0.494$ (0.149), $\beta_m = -0.026$ (0.095)
- (b) SHAR: $\beta^+ = 0.401$ (0.252), $\beta^- = 0.324$ (0.236); difference -0.078 , $t = -0.17$, one-sided $p = 0.566$: do not reject H_0
- (c) HARQ: $\hat{\beta}_Q = -35.062$ (15.904), $t = -2.20 < -1.645$: reject H_0 at 5% one-sided

B3 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: the two SHAR intervals are wide and overlap (the semivariances are highly correlated for one stock); the HARQ interval lies below zero
- Interpretation: on noisy days HARQ trusts the day's RV less, and this holds for JPM; the good/bad asymmetry is a panel result [Patton & Sheppard \(2015\)](#), not detectable for JPM alone



- SHAR beta+ (positive semivariance)
- SHAR beta- (negative semivariance)
- HARQ beta_Q

B5 [Solved]: Sig-LK vs log-HAR for JPM, One Day Ahead — Task

- **Question:** does the signature method of the lecture (Sig-LK) forecast JPM's next-day RV better than log-HAR?
- **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; five-minute RV, daily open and close prices
 - ▶ Design of [Chapter 20](#) (primer slide on Sig-LK): rolling window of 1000 days, re-estimation every 5 days, path (time, ln RV, cumulative r) over 22 days, depth 3, two-step LASSO with BIC, kernel temperature $c = 0.5$
- **Tasks**
 - (a) Produce one-day forecasts of Sig-LK and log-HAR from 27 Jan 2020 to 29 Jan 2026 (1511 days) and compute the mean QLIKE of each.
 - (b) Compute the ratio of the two mean losses and its 95% CI with the stationary bootstrap (mean block length 10, 1000 replications).
 - (c) Run the DM test (HAC variance, bandwidth $\max(h, \lceil T^{1/3} \rceil)$) and the conditional GW test with instruments $(1, d_{t-1})$ [Giacomini & White \(2006\)](#).
 - (d) Plot the cumulative loss difference $\sum_{s \leq t} d_s$.
- **Report:** losses, ratio and CI, t , p , decisions, the chart
- **Interpretation:** is there any evidence that the signature adds value for JPM?
- **Notebook:** section B5

B5 [Solved]: Solution (1/2) — Losses and Tests

- (a) Mean QLIKE: Sig-LK 0.1726, log-HAR 0.1727; on average 4.2 regressors selected, effective sample size 851 of 1000
- (b) Ratio 1.000; bootstrap 95% CI [0.956, 1.047] contains 1
- (c) DM: $d_t = \text{QLIKE}_t^{\text{Sig-LK}} - \text{QLIKE}_t^{\text{log-HAR}}$, $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}} / T} = -0.02$, one-sided $p = 0.493$: do not reject equal accuracy
- GW: Wald statistic $2.85 \sim \chi^2(2)$, $p = 0.241$: the instruments $(1, d_{t-1})$ give no significant evidence that the loss difference is predictable

B5 [Solved]: Solution (2/2) — Cumulative Loss Difference

- (d) Upward segments: log-HAR better; downward segments: Sig-LK better; the largest moves sit in March 2020 and April 2025
- A handful of stress days decides the sign of the total: report the cumulative path, not only the mean
- Interpretation: no evidence; the two forecasts are practically identical, and the CI rules out gains or losses larger than about 5%



— Cumulative QLIKE(Sig-LK) - QLIKE(log-HAR), JPM, $h = 1$

B7 [Solved]: Holm and BH, Step by Step — Task

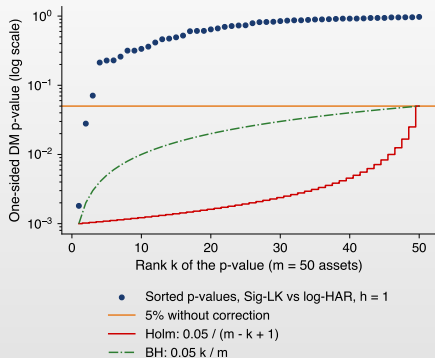
- **Question:** after correcting for 50 tests, on how many assets does Sig-LK beat log-HAR one day ahead?
- **Inputs**
 - ▶ Per-asset DM p -values from Lecture 20 (provided with the notebook): one-sided DM p -values for “Sig-LK better than log-HAR” at $h = 1$ (QLIKE), one per asset; family = the 50 assets
 - ▶ The five FX pairs, sorted: EURUSD 0.071, USDJPY 0.213, USDCAD 0.336, AUDUSD 0.718, GBPUSD 0.817
 - ▶ Holm (FWER 5%): compare the k -th smallest $p_{(k)}$ with $0.05/(m - k + 1)$ and stop at the first non-rejection; BH (FDR 5%) [Benjamini & Hochberg \(1995\)](#): reject the k smallest for the largest k with $p_{(k)} \leq 0.05 k/m$
- **Tasks**
 - (a) Apply Holm to the five FX p -values ($m = 5$).
 - (b) Apply BH to the same five.
 - (c) Repeat both procedures for all 50 assets, count the rejections and plot the sorted p -values against the three thresholds.
- **Report:** rejections raw, Holm, BH; the chart
- **Interpretation**
 - ▶ what may we claim?
 - ▶ about how many assets?
- Notebook: section B7

B7 [Solved]: Solution (1/2) — Steps

- (a) Holm, $m = 5$: $p_{(1)} = 0.071$ vs $0.05/5 = 0.01$: no rejection, stop; Holm-adjusted p : 0.354, 0.852, then 1
- (b) BH: thresholds 0.01, 0.02, 0.03, 0.04, 0.05; every $p_{(k)}$ is above its threshold: no rejection; the smallest BH-adjusted p is 0.354
- (c) 50 assets: 2 raw rejections; Holm: the smallest, ES with $p = 0.002$, is above $0.05/50 = 0.001$: stop, 0 rejections; BH: $p_{(1)} > 0.001$ and $p_{(2)} = 0.028 > 0.002$, no k qualifies: 0

B7 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: only the lowest points cross the 5% line; none reaches the Holm staircase or the BH line
- Holm is valid under any dependence; BH needs independent or positively dependent (PRDS) p -values [Benjamini & Yekutieli \(2001\)](#)
- Interpretation: no asset-level improvement survives either correction; ES is at most a lead for a new, pre-registered study



C2 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: “rv holds the daily 5-minute RV of one stock. Build the signature LASSO of Gu et al. for next-day log RV and evaluate it with QLIKE on the last 30% of the sample.”

- **Answer:** the code below and the conclusion

```

lrv = np.log(rv.values)
S = np.array([sig_path(lrv[t-21:t+1, None], depth=3)      # 22-day window
              for t in range(21, len(lrv) - 1)])
y = lrv[22:]                                              # next-day log RV
lam = LassoCV(cv=5).fit(S, y).alpha_                     # choose lambda
cut = int(0.7 * len(y))
m = Lasso(alpha=lam).fit(S[:cut], y[:cut])
f = np.exp(m.predict(S[cut:]))
qlike = lambda a, b: a / b - np.log(a / b) - 1
print(qlike(f, np.exp(y[cut:])).mean())

```

- **Conclusion of the AI:** “The signature LASSO has a lower QLIKE than HAR on the test set, so path signatures capture the shape of volatility and improve the forecast.”

C2 [Proposed]: Critique an AI Answer — Task

Solution discussed in class

- **Question:** which errors in the answer would change the conclusion?
- **Model:** A1 (time augmentation), A7 (QLIKE), B5 (evaluation without look-ahead); the code contains **at least three** errors
- **Tasks**
 - (a) Find at least three errors; for each, name the line and explain in one sentence why it is wrong.
 - (b) For each error, describe a check that reveals it without knowing the answer (a unit test, a timing test, a known case).
 - (c) Assess the exponential back-transform and whether the code is the two-step weighted method of Gu et al.
 - (d) Write the corrected code and compare it with log-HAR on the same split, with a DM test.
 - (e) Log each error as in [AI_ERRORS.md](#): request, answer, error, how found, correction.
- **Interpretation:** does the conclusion follow even if the code were correct?
- AI code usually runs without an error message: the mistakes are in the method, not in the syntax

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Signature algebra	A2, A3 [Solved], A4	A1
2. HAR as an AR(22)	A5 [Solved], A6	B3, Wald tests
3. Losses and weights	A8, A10	A7, A9
4. Other assets	B2 (ES), B4 (CL), B6 (ES, $h = 5$)	B1, B3, B5
5. Multiple testing at one month	B8	B7
6. Project seed	C1, a level-aware kernel	B5, Lecture 20 (kernel weights)

- Each exercise has its own notebook section with starter code (an empty code cell for proposed ones); within a pack, work in the order given
- Proposed exercises name their model: the solved exercise that uses the same technique

A2 [Proposed]: Signature and Shuffle Identity — Task

Solution discussed in class

▣ Questions

- ▶ what is the signature of a three-segment path in the plane?
- ▶ which channel moved first?

▣ Model: A1

- ▣ **Inputs:** A path in $\mathbb{R}^2$ through $(0, 0) \rightarrow (2, 1) \rightarrow (3, -1) \rightarrow (4, 2)$, linear between points

▣ Tasks

- (a) Write the increments and compute level 1.
- (b) Compute level 2 with the formula of A1 (squares plus ordered pairs).
- (c) Verify $S^1 S^2 = S^{12} + S^{21}$ and compute the Lévy area A^{12} .

- ▣ **Report:** $S^{(1)}$, $S^{(2)}$, A^{12} ; **Interpretation:** which channel moved first, on balance?

- ▣ **Notebook:** section A2

A3 [Solved]: Chen's Identity and Path Reversal — Task

▣ Questions

- ▶ what is the signature of a path run there and back?
- ▶ why does this matter for time augmentation?

▣ Inputs

- ▶ Path $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2)$ from the lecture, with $S^{(1)} = (1, 2)$ and $S^{(2)} = \begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix}$
- ▶ $\overleftarrow{X}$ is the same path run backwards: $(1, 2) \rightarrow (1, 0) \rightarrow (0, 0)$

▣ Tasks

- (a) Compute levels 1 and 2 of $\text{Sig}(\overleftarrow{X})$ with the A1 formula.
 - (b) Apply Chen's identity, $(A \otimes B)^{(2)} = A^{(2)} + A^{(1)} B^{(1)\top} + B^{(2)}$, to compute level 2 of $X * \overleftarrow{X}$.
 - (c) State what the result implies for the uniqueness of the signature.
- ▣ **Report:** the two signatures and level 2 of the concatenation; **Interpretation:** why do we add a time channel in practice?
 - ▣ **Notebook:** section A3

A3 [Solved]: Solution

- (a) Reversed increments $(0, -2), (-1, 0)$: level 1 is $(-1, -2) = -S^{(1)}$; level 2: squares $\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$ plus the pair $\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}$, so $\overleftarrow{S}^{(2)} = \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix}$
- (b) Chen: $\begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$; level 1 is $S^{(1)} - S^{(1)} = 0$ as well
- (c) There and back is “tree-like”: its signature is that of the constant path; the signature identifies a path only up to such pieces [Hambly & Lyons \(2010\)](#)
- Interpretation: a strictly increasing time channel cannot go back, so tree-like pieces are impossible and the signature determines the path up to its starting point: we time-augment and supply the level separately

A4 [Proposed]: Lévy Area of a Loop — Task

Solution discussed in class

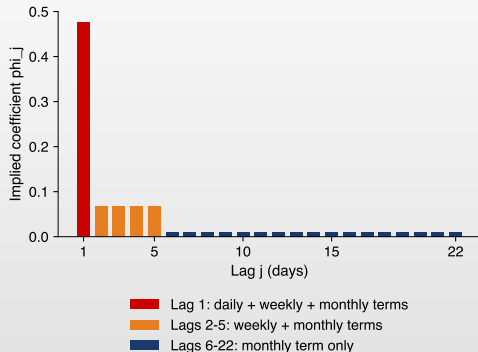
- ▣ **Question:** what does level 2 of the signature see in a closed loop that level 1 misses?
- ▣ **Model:** A3
- ▣ **Inputs:** The unit square $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 0)$, four linear segments
- ▣ **Tasks**
 - (a) Compute level 1 and the Lévy area A^{12} for the square run anticlockwise.
 - (b) Repeat for the square run clockwise, and for the square translated by $(5, 5)$.
 - (c) Explain why level 1 cannot distinguish the loop from a constant path while level 2 can.
- ▣ **Report:** three values of $S^{(1)}$ and A^{12} ; **Interpretation:** which invariance does the translation illustrate?
- ▣ **Notebook:** section A4

A5 [Solved]: HAR as a Restricted AR(22) — Task

- ▣ **Question:** which 22 lag coefficients does a HAR model for $\ln RV$ imply for JPM?
- ▣ **Inputs**
 - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only
 - ▶ HAR with *averages of logs*: $\ell_t = \ln RV_t$, $\ell_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 \ell_{t-j}$, $\ell_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} \ell_{t-j}$
 - ▶ The lecture's log-HAR uses logs of averages, $\ln RV_t^{(w)} \neq \ell_t^{(w)}$ (Jensen), and has no exact AR form
 - ▶ Full sample (2764 days), OLS, Newey–West (5 lags) s.e.: $\hat{\beta}_d = 0.408$ (0.027), $\hat{\beta}_w = 0.299$ (0.046), $\hat{\beta}_m = 0.196$ (0.040)
- ▣ **Tasks**
 - (a) Substitute $\ell_t^{(w)}$ and $\ell_t^{(m)}$ into the HAR equation and collect the coefficient ϕ_j of each lag ℓ_{t-j+1} , $j = 1, \dots, 22$.
 - (b) Compute $\phi_1, \phi_2 = \dots = \phi_5$ and $\phi_6 = \dots = \phi_{22}$ from the JPM estimates and plot them.
 - (c) Compute the sum $\sum_{j=1}^{22} \phi_j$.
- ▣ **Report:** the three distinct values, the sum, the chart
- ▣ **Interpretation**
 - ▶ what does the sum measure?
 - ▶ is the process stationary?
- ▣ Notebook: section A5

A5 [Solved]: Solution

- (a) $\beta_w \ell_t^{(w)} = \sum_{j=1}^5 (\beta_w/5) \ell_{t-j+1}$ and $\beta_m \ell_t^{(m)} = \sum_{j=1}^{22} (\beta_m/22) \ell_{t-j+1}$: lag 1 collects all three terms, lags 2–5 the weekly and monthly ones, lags 6–22 only the monthly one
- (b) $\phi_1 = 0.408 + 0.299/5 + 0.196/22 = 0.476$; $\phi_{2..5} = 0.0688$; $\phi_{6..22} = 0.0089$
- (c) $\sum_j \phi_j = \beta_d + \beta_w + \beta_m = 0.903$
- Interpretation: the sum is the persistence of log RV; all $\phi_j \geq 0$ and the sum is below 1, so the process is stationary, with a slow, long-memory-like decay



A6 [Proposed]: Testing the HAR Restrictions — Task

Solution discussed in class

- ▣ **Question:** how do we test the restrictions that HAR imposes on an unrestricted AR(22)?
- ▣ **Model:** A5
- ▣ **Inputs**
 - ▶ The unrestricted AR(22) for $\ell_t = \ln RV_t$ has an intercept and slopes $\phi_1, \dots, \phi_{22}$; HAR with averages of logs (A5) is its restricted version
 - ▶ ϕ is the linear projection of ℓ_{t+1} on its 22 lags; weakly dependent data with finite fourth moments; errors may be heteroskedastic and serially correlated
- ▣ **Tasks**
 - (a) Count the linear restrictions and write them as $R\phi = 0$.
 - (b) Write the Wald statistic with an HAC covariance and its distribution under H_0 .
 - (c) State the 5% critical value.
- ▣ **Interpretation:** if the test rejects in a long sample, should we drop HAR for forecasting?
- ▣ **Notebook:** section A6

A8 [Proposed]: QLIKE With Its Arguments Swapped — Task

Solution discussed in class

- ▣ **Question:** which forecasts does a QLIKE with swapped arguments reward?
- ▣ **Model:** A7
- ▣ **Inputs:** A program computes $L(RV, F) = F/RV - \ln(F/RV) - 1$, QLIKE with the arguments swapped
- ▣ **Tasks**
 - (a) Find the forecast $F^\dagger$ that minimises $\mathbb{E}_t[L(RV, F)]$.
 - (b) Compare $F^\dagger$ with $\mathbb{E}_t[RV]$ using Jensen's inequality.
 - (c) Say which models the swapped loss favours.
- ▣ **Interpretation:** why can this bug pass elementary checks?
- ▣ **Notebook:** section A8

A10 [Proposed]: Limits of the Temperature — Task

Solution discussed in class

▣ Questions

- ▶ what happens to the kernel weights at the two extreme temperatures?
- ▶ why scale by the median distance?

▣ Model: A9

- ▣ **Inputs:** n past windows, distances $\delta_\tau > 0$, temperature $\gamma = c/\text{median}(\delta)$ as in the lecture

▣ Tasks

- (a) Find the limits of w and n_{eff} when $c \rightarrow 0$.
- (b) Find the limits when $c \rightarrow \infty$ (assume a unique nearest window).
- (c) Show that multiplying all distances by $k > 0$ leaves the weights unchanged.

- ▣ **Interpretation:** why is the median scaling needed to compare c across 50 assets?

- ▣ Notebook: section A10

B2 [Proposed]: The Lévy Area for E-mini S&P 500 Futures — Task

Solution discussed in class

- ▣ **Question:** does the Lévy area predict next month's volatility of E-mini S&P 500 futures (ES)?
- ▣ **Model:** B1
- ▣ **Inputs:** VOLARE futures data, 28 Sep 2009 – 30 Jan 2026, ES; five-minute RV, open and close prices; target, regressors and 22 Newey–West lags as in B1
- ▣ **Tasks**
 - (a) Repeat steps (a)–(d) of B1 for ES.
 - (b) Test the coefficient of A_t at 1% and 5%, two-sided.
 - (c) Compare the R^2 gain with the one for JPM.
- ▣ **Interpretation**
 - ▶ what does a positive coefficient say about windows with a negative area?
 - ▶ can the sign alone label them leverage episodes?
- ▣ **Notebook:** section B2

B4 [Proposed]: HAR, SHAR and HARQ for Crude Oil — Task

Solution discussed in class

- ▣ **Question:** do the SHAR asymmetry and the HARQ effect hold for crude oil futures (CL)?
- ▣ **Model:** B3
- ▣ **Inputs:** VOLARE futures data, 28 Sep 2009 – 30 Jan 2026, CL; five-minute RV, semivariances and realised quarticity; models, tests and Newey–West (5 lags) as in B3
- ▣ **Tasks**
 - (a) Estimate SHAR and test $\beta^- > \beta^+$ as in B3(b).
 - (b) Estimate HARQ and test $\beta_Q < 0$ as in B3(c); plot the coefficients with 95% HAC intervals.
 - (c) Compare both results with JPM.
- ▣ **Interpretation:** why might the asymmetry be stronger for a commodity than for a bank stock?
- ▣ **Notebook:** section B4

B6 [Proposed]: Sig-LK vs log-HAR for ES, One Week Ahead — Task

Solution discussed in class

- ▣ **Question:** does Sig-LK beat log-HAR for ES when the target is the average RV of the next five days?
- ▣ **Model:** B5
- ▣ **Inputs:** VOLARE futures data, 28 Sep 2009 – 30 Jan 2026, ES; design of B5 with $h = 5$ and $c = 1.0$ (futures validation block)
- ▣ **Tasks**
 - (a) Produce weekly forecasts of Sig-LK and log-HAR with the design of B5.
 - (b) Compute the QLIKE ratio and its stationary-bootstrap 95% CI (mean block length 10).
 - (c) Run the DM test (bandwidth at least $h = 5$) and the conditional GW test; plot the cumulative loss difference.
- ▣ **Interpretation**
 - ▶ why is the CI so asymmetric?
 - ▶ what does it imply for the power of DM?
- ▣ **Notebook:** section B6

B8 [Proposed]: Multiple Testing at One Month Against HAR — Task

Solution discussed in class

- ▣ **Question:** on how many assets does Sig-LK beat HAR one month ahead after correcting for 50 tests?
- ▣ **Model:** B7
- ▣ **Inputs:** Lecture 20 results: 50 one-sided DM p -values for “Sig-LK better than HAR” at $h = 22$ (QLIKE)
- ▣ **Tasks**
 - (a) Count the rejections at 5% without correction.
 - (b) Apply Holm and BH as in B7, count the rejections and name the asset with the smallest p -value.
 - (c) Compare with the pooled mean QLIKE ratio of Sig-LK relative to HAR at $h = 22$ (1.334).
- ▣ **Interpretation:** how can Sig-LK beat HAR on several assets and still lose on average?
- ▣ **Notebook:** section B8

C1 [Proposed]: A Level-Aware Signature Kernel — Task

Solution discussed in class

- ▣ **Question:** does a kernel that also sees the *level* of volatility make the weights useful in a crash?
- ▣ **Inputs**
 - ▶ The kernel of the lecture is translation-invariant, so it cannot recognise a high-volatility regime (Lecture 20, weights on 16 March 2020); data: VOLARE, the 40 stocks, 2015–2026
 - ▶ Idea: prepend a **basepoint** [Morrill et al. \(2020\)](#): the path starts at the training mean of $\ln RV$ and jumps to the first value of the window; by Chen, $\text{Sig} = \exp_{\otimes}(v) \otimes \text{Sig}(X)$ with v the standardised level jump
- ▣ **Tasks**
 - (a) Write a one-page pre-registration: 40 stocks, $h = 1$, the lecture design, $c = 0.5$, QLIKE, DM with Holm, the COVID-19 subperiod reported separately.
 - (b) Implement the basepoint kernel (use the Chen update of the notebook) with the same features as Sig-LK, and forecast.
 - (c) Compare with Sig-LK and log-HAR: mean QLIKE ratios (all days and COVID-19), wins over log-HAR raw and after Holm.
- ▣ **Interpretation:** if seeing the level does not help, where would you look next (features, target, horizon)?
- ▣ Notebook: section C1

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20; data: volare.unime.it

References

- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 1165–1188.
- Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18.
- Chen, K.-T. (1957). Integration of paths, geometric invariants and a generalized Baker–Hausdorff formula. *Annals of Mathematics*, 65(1), 163–178.
- Chevyrev, I., Kormilitzin, A. (2016). A primer on the signature method in machine learning. arXiv:1603.03788.
- Cipollini, F., Cruciani, G., Gallo, G.M., Insana, A., Otranto, E., Spagnolo, F. (2026). VOLatility Archive for Realized Estimates (VOLARE). arXiv:2602.19732.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gu, H., Guo, X., Jacobs, T.L., Kaminsky, P., Li, X. (2024). Transportation marketplace rate forecast using signature transform. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4997–5005.
- Hambly, B., Lyons, T. (2010). Uniqueness for the signature of a path of bounded variation and the reduced path group. *Annals of Mathematics*, 171(1), 109–167.
- Morrill, J., Fermanian, A., Kidger, P., Lyons, T. (2020). A generalised signature method for multivariate time series feature extraction. arXiv:2006.00873.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- Patton, A.J., Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3), 683–697.